

Safeband_X – ESP32-CAM Pin Connection Summary

Project: Safeband_X – IoT Personal Safety Device

Controller: AI Thinker ESP32-CAM

Buttons

- Panic Button (3s hold): GPIO 0 → Button → GND (INPUT_PULLUP)
- Safe Cancel Button: GPIO 4 → Button → GND (INPUT_PULLUP)

Alert Outputs

- Piezo Siren: GPIO33 → 1kΩ → Transistor Base, Emitter → GND, Collector → Siren -, Siren + → 5V
- Red LED: GPIO32 → 470Ω → LED Anode, LED Cathode → GND

GPS Module (Neo-6M)

- VCC → 5V
- GND → GND
- TX → GPIO 3 (U0R)
- RX → GPIO 1 (U0T)
- Baud Rate: 9600 (TinyGPS++)

MPU6050 (Optional Impact Sensor)

- VCC → 3.3V
- GND → GND
- SDA → GPIO 13
- SCL → GPIO 15
- INT → GPIO 12

Camera & Storage

- OV2640 Camera: Built-in (no wiring)
- MicroSD Card: Built-in SD_MMC slot

Programming / Power (FTDI)

- 5V → 5V
- GND → GND
- TX → GPIO 3
- RX → GPIO 1
- GPIO0 → GND (upload only)

Design Note

- INMP441 audio not connected due to SD card pin conflicts (GPIO 2, 14, 15).
- Requires external SD or different ESP32 board for audio + SD.